Alejandro de la Fuente

Melbourne, VIC | Australian PR | alejandrofuentepinero@gmail.com | 0499 501 813

LinkedIn | GitHub | Portfolio

AI ENGINEER

SUMMARY

Quantitative scientist building end-to-end LLM systems: data pipelines at scale, advanced RAG with rigorous evaluation, fine-tuning across open-source and frontier models, and autonomous agents. Six years of research-grade modelling bring evaluation discipline, systems-level reasoning, and rigorous handling of uncertainty, foundations that trustworthy LLM systems depend on.

CORE SKILLS

- LLM engineering: prompt engineering, RAG (LangChain, ChromaDB, vector databases, embeddings), fine-tuning (QLoRA, frontier SFT via OpenAI and Anthropic APIs), tool-calling, multi-agent orchestration, structured outputs (Pydantic), LangGraph
- Evaluation: retrieval metrics (MRR, nDCG), LLM-as-judge, benchmarking, validation and calibration discipline
- Engineering and deployment: Python (pandas, NumPy, scikit-learn, XGBoost, PyTorch), SQL (PostgreSQL), R, Git/GitHub, serverless on Modal, Weights & Biases, Gradio, Streamlit, AWS Certified Cloud Practitioner
- ML and statistics: Bayesian hierarchical and spatiotemporal modelling, supervised and unsupervised learning, NLP, ranking and recommendation, A/B testing, causal inference

FEATURED PROJECTS

LLM Engineering Lab (GitHub) | Python, LangChain, ChromaDB, QLoRA, multi-agent, Modal

- Built an autonomous price intelligence system that ingests 820k Amazon products, learns what each item should cost via an ensemble of QLoRA-fine-tuned Llama-3.2-3B, frontier-tuned GPT-4.1, and a RAG layer over 800k vectors (benchmarked across a dozen model families), then deploys agents that scan live deal feeds, flag underpriced items, and push real-time notifications.
- Built an advanced RAG with hierarchical retrieval, query rewriting, and LLM reranking, paired with an evaluation harness (MRR, nDCG, LLM-as-judge) that quantifies the lift across architectural changes.

AI-JIE: LLM Extraction & Evaluation Pipeline (**GitHub**) | Python, OpenAI API, Pydantic, HuggingFace Hub

- Solved a hard extraction problem most pipelines get wrong: reliably distinguishing required, preferred, and soft skills in raw job postings. Async LLM pipeline with chain-of-thought scaffolding and deterministic post-processing, calibrated against human review (4.11/5) and shipped as a published HuggingFace dataset.

Job Intelligence Engine (Live App | GitHub) | Python, XGBoost/LightGBM, SBERT, NetworkX, Streamlit, SHAP

- Built a job recommender that models the job market as a system, capturing how roles, skills, and demand interact, to surface best-now and stretch shortlists and design dynamic upskilling paths. Trained on 6,100+ real postings with SHAP-based explanations and counterfactual upskilling lift; deployed as a live app.

EXPERIENCE

Postdoctoral Fellow (Forecasting & Risk Modelling) | James Cook University | Sep 2024–Present

Develop forecasting and risk models delivered as a decision-support tool for prioritising intervention sites, analysing ~15,000 records across 28 years and 500+ locations..

Research Assistant (Data Analytics) | University of Melbourne | 2024

Built and deployed reproducible data pipelines processing 27,000 records across multiple sources; delivered analysis-ready datasets and stakeholder reporting on a defined timeline.

PhD Researcher (Bayesian Spatiotemporal Modelling) | James Cook University | Aug 2020–Jun 2024

Developed Bayesian hierarchical and spatiotemporal forecasting models on 30-year monitoring data across 150+ locations to estimate trends under climate extremes, producing uncertainty-aware evidence that supported protection-listing decisions for 15 rainforest species.

EDUCATION

PhD (Quantitative Ecology), Cum Laude | James Cook University, 2024

MSc (Biodiversity Conservation, quantitative focus) | University of Salamanca, 2016

Publications: First-author work in top-tier journals on Bayesian modelling and climate-risk inference